

HYOJEONG SON

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Academic Positions

2026 – present **Postdoctoral Fellow in Mathematics**, University of British Columbia, Vancouver
Mentors: Omer Angel and Jonathan Hermon

Education

2021 – 2026 **Ph.D.** in Mathematics, University of Washington (UW)
Advisor: Christopher Hoffman

2019 – 2021 **A.M.** in Mathematics, Washington University in St. Louis (WashU)
Advisor: Renato Feres

2015 – 2018 **B.S.**, Stony Brook University
B.S. in Mathematics and Applied Mathematics & Statistics; minor in Computer Science

Publications

2026 Christopher Hoffman, Avi Levy, Elchanan Mossel, and Hyojeong Son.
Stable Matchings with Correlated Preferences (in preparation).

2026 Christopher Hoffman, Jacob Richey, and Hyojeong Son.
Local Density of Activated Random Walk on $\mathbb{Z}$.

2024 Madeline Brown, Christopher Hoffman, and Hyojeong Son.
Supercritical Activated Random Walk on $\mathbb{Z}$ is Explosive.

2021 Robert Hough and Hyojeong Son.
Cut-off for Sandpiles on Tiling Graphs. **Annals of Probability** 49(2), 671–731.

2021 Robert Hough and Hyojeong Son.
The Spectrum of the Abelian Sandpile Model. **Mathematics of Computation** 90(327), 441–469.

Awards & Honors

2025 Excellence in Teaching
Department of Mathematics, University of Washington

2024 Lisa Tanzi and Charles Egerton Endowed Graduate Fellowship
Department of Mathematics, University of Washington

2018 Undergraduate Recognition Award, Academic Excellence
Stony Brook University

Invited Talks

February 2026 UW Probability Seminar, Seattle, WA
Local density of point-source activated random walk on $\mathbb{Z}$

June 2025 CRM–PIMS Summer School in Probability, Vancouver, Canada
Self-Organized Criticality

March 2025 Cornell Probability Seminar, Ithaca, NY
Self-Organized Criticality

February 2025 University of Washington Graduate Student Analysis Seminar, Seattle, WA
Stable Matching

- December 2024 University of Washington Graduate Student Analysis Seminar, Seattle, WA
Universality for Self-Organized Criticality
- September 2024 University of Washington General Examination Talk, Seattle, WA
Stable Matching

Mentorship

- Winter 2025 – present Washington Experimental Mathematics Lab (WXML), UW
Mentoring undergraduates estimating critical exponents for activated random walks on various lattices
- Spring 2024 Washington Experimental Mathematics Lab (WXML), UW
Mentored undergraduates on projects involving voting data, map analysis, and ecological inference in the Yakima Valley
- Spring 2021 Graduate Student-Led Reading Program, WashU
Led a reading group for mathematics undergraduates focused on sandpiles

Teaching

Instructor, University of Washington

Math 394 Probability I
Summer 2025

Teaching Assistant, University of Washington

Math 111 Algebra
Fall 2022

Math 125 Single Variable Calculus
Fall 2021, Winter 2022, Fall 2023

Math 126 Multivariable Calculus
Spring 2022, Summer 2022

Math 207 Differential Equations
Winter 2023, Spring 2023

Math 394 Probability I
Winter 2024, Fall 2024, Spring 2025

Math 395 Probability II
Spring 2024, Winter 2025

Math 491 Stochastic Processes and Option Pricing
Fall 2025

Math 524 Graduate Real Analysis I
Winter 2026

Math 525 Graduate Analysis II
Spring 2026